

Chapter 1

Preliminaries

References

1. Mood et al. (2010): Preliminaries
2. Hogg et al. (2019): Preliminaries
3. Ali (1973): Preliminaries
4. Gupta and Kapoor (2000): Preliminaries, χ^2 , t , F —distributions
5. Sahoo (2013): χ^2 , t , F —distributions, Fisher's- z
6. Wikipedia: Central χ^2 , t , F —distributions, Non-central χ^2 , t , F —distributions, More on Sampling Distributions.

1.1 Statistic Versus Parameter

In order to avoid verbal confusion with the statistical constants of the population, viz., mean (μ), variance (σ^2) which are usually referred to as parameters, statistical measures computed from the sample observations alone, e.g., mean ($\bar{x}$), variance (s^2), etc., have been termed by Professor R.A. Fisher as statistics.

In practice parameter values are not known and the estimates based on the sample values are generally used. Thus statistic which may be regarded as an estimate of parameter, obtained from the sample, is a function of the sample values only. It may be pointed out that a statistic, as it is based on sample values and as there are

multiple choices of the samples that can be drawn from a population, varies from sample to sample. The determination or the characterization of the variation (in the values of the statistic obtained from different samples) that may be attributed to chance or fluctuations of sampling is one of the fundamental problems of the sampling theory.

1.2 Estimator Versus Estimate

An estimator is a rule for calculating an estimate of a given quantity based on observed data: thus the rule (the estimator), and its result (the estimate) are distinguished. The sample variance is an estimator of the population variance.

1.3 Estimator Versus Statistic

A statistic is a function of a sample, whereas an estimator is a function of a sample related to some quantity of the distribution. An estimator is a statistic (that is, a function of the data) that is used to infer the value of an unknown parameter in a statistical model.

1.4 Standard Errors

The standard error ($S.E$) of a statistic (usually an estimate of a parameter) is the standard deviation of its sampling distribution or an estimate of that standard deviation. If the parameter or the statistic is the mean, it is called the standard error of the mean (SEM). *So, the standard deviation of the sampling distribution of a statistic is known as its Standard Error, abbreviated as $S.E$.*

1.5 Sampling Distribution Versus Parent Distribution

The sampling distribution of a statistic is the distribution of that statistic, considered as a random variable, when derived from a random sample of size n . It may be considered as the distribution of the statistic for all possible samples from

the same population of a given sample size. *The sampling distribution depends on the underlying distribution of the population, the statistic being considered, the sampling procedure employed, and the sample size used.*

For example, consider a normal population with mean μ and variance σ^2 . Assume we repeatedly take samples of a given size from this population and calculate the arithmetic mean $\bar{x}$ for each sample, this statistic is called the sample mean. The distribution of these means, or averages, is called the “sampling distribution of the sample mean”. This distribution is normal $N(\mu, \sigma^2/n)$ (n is the sample size) since the underlying population is normal, although sampling distributions may also often be close to normal even when the population distribution is not normal (see central limit theorem). An alternative to the sample mean is the sample median. When calculated from the same population, it has a different sampling distribution to that of the mean and is generally not normal (but it may be close to normal for large sample sizes).

1.5.1 Sampling Distribution of a Statistic

If we draw a sample of size n from a given finite population of size N , then the total number of possible samples is:

$${}^N C_n = \frac{N!}{n! (N - n)!} = k, (\text{say})$$

For each of these k samples we can compute some statistic $t = t(x_1, x_2, \dots, x_n)$, in particular the mean $\bar{x}$, the variance s^2 , etc., as given below:

The set of the values of the statistic so obtained, one for each sample, constitutes what is called the sampling distribution of the statistic. For example, the values $t_1, t_2, t_3, \dots, t_k$ determine the sampling distribution of the statistic t . In other words, statistic t may be regarded as a random variable which can take the values $t_1, t_2, t_3, \dots, t_k$ and we can compute the various statistical constants like mean,

Table 1.1: Sampling distribution of a statistic

Sample Number	Statistic		
	t	$\bar{x}$	s^2
1	t_1	$\bar{x}_1$	s_1^2
2	t_2	$\bar{x}_2$	s_2^2
3	t_3	$\bar{x}_3$	s_3^2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	t_k	$\bar{x}_k$	s_k^2

variance, skewness, kurtosis etc., for its distribution. For example, the mean and variance of the sampling distribution of the statistic t are given by:

$$\bar{t} = \frac{1}{k} \sum_{i=1}^k t_i,$$

and

$$Var(t) = \frac{1}{k} \sum_{i=1}^k (t_i - \bar{t})^2.$$

1.5.1.1 Sampling With Versus Without Replacement

Example Let $N = (1, 2, 3)$. Draw a sample of size $n = 2$ with and without replacement by considering order of the sample.

Solution:

Ordered sampling with replacement

(1,1), (1,2), (1,3)

(2,1), (2,2), (2,3)

(3,1), (3,2), (3,3)

So, the number of possible samples = $N^n = 3^2 = 9$.

Ordered sampling without replacement

(1,2), (1,3)

(2,1), (2,3)

(3,1), (3,2)

So, the number of possible samples = $\frac{N!}{(N-n)!} = \frac{3!}{1!} = 6$.

Unordered sampling with replacement

(1,1), (1,2), (1,3)

(2,2), (2,3)

(3,3)

So, the number of possible samples = $\frac{(N+n-1)!}{(n)!(N-1)!} = \frac{4!}{2!2!} = 6$.

Unordered sampling without replacement

(1,2), (1,3)

(2,3)

So, the number of possible samples = $\frac{(N)!}{(n)!(N-n)!} = \frac{3!}{2!1!} = 3$.

Example 1

Draw all possible samples of size 2 without replacement from a population consisting of 3, 6, 9, 12, 15. Form the sampling distribution of sample means and verify the following results:

(i) $E(\bar{X}) = \mu,$

(ii) $Var(\bar{X}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right).$

Solution: We have population values 3, 6, 9, 12, 15, population size $N = 5$ and sample size $n = 2$. Thus, the number of possible samples which can be drawn without replacement is

$$\binom{N}{n} = \binom{5}{2} = \frac{5!}{3! 2!} = 10$$

Table 1.2: Numerical example for sampling distribution of a statistic

Sample Number	Sample Values	Sample Mean ($\bar{X}$)
1	3, 6	4.5
2	3, 9	6
3	3, 12	7.5
4	3, 15	9
5	6, 9	7.5
6	6, 12	9
7	6, 15	10.5
8	9, 12	10.5
9	9, 15	12
10	12, 15	13.5

Hence,

$$\begin{aligned} E(\bar{X}) &= \sum \bar{X} f(\bar{X}) \\ &= \frac{90}{10} = 9, \end{aligned}$$

Table 1.3: Sampling distribution of the sample mean $\bar{X}$

$\bar{X}$	f	$f(\bar{X})$	$\bar{X}f(\bar{X})$	$\bar{X}^2f(\bar{X})$
4.5	1	$\frac{1}{10}$	$\frac{4.5}{10}$	$\frac{20.25}{10}$
6.0	1	$\frac{1}{10}$	$\frac{6.0}{10}$	$\frac{36.00}{10}$
7.5	2	$\frac{2}{10}$	$\frac{15.0}{10}$	$\frac{112.50}{10}$
9.0	2	$\frac{2}{10}$	$\frac{18.0}{10}$	$\frac{162.00}{10}$
10.5	2	$\frac{2}{10}$	$\frac{21.0}{10}$	$\frac{220.50}{10}$
12.0	1	$\frac{1}{10}$	$\frac{12.0}{10}$	$\frac{144.00}{10}$
13.5	1	$\frac{1}{10}$	$\frac{13.5}{10}$	$\frac{182.25}{10}$
Total	10	1	$\frac{90}{10}$	$\frac{877.50}{10}$

$$\begin{aligned}
 Var(\bar{X}) &= \sum \bar{X}^2 f(\bar{X}) - \left[\sum \bar{X} f(\bar{X}) \right]^2 \\
 &= \frac{877.50}{10} - 9^2 = 87.75 - 81 = 6.75.
 \end{aligned}$$

The mean and variance of the population are:

X	3	6	9	12	15	$\sum X = 45$
X^2	9	36	81	144	225	$\sum X^2 = 495$

$$\mu = \frac{\sum X}{n} = 9$$

and

$$\begin{aligned}
 \sigma^2 &= \frac{\sum X^2}{N} - \left(\frac{\sum X}{n} \right)^2 \\
 &= \frac{495}{5} - 81 = 99 - 81 = 18.
 \end{aligned}$$

Verification:

(i) $E(\bar{X}) = \mu = 9,$

(ii) $Var(\bar{X}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = \frac{18}{2} \left(\frac{5-2}{5-1} \right) = 6.75.$

Assignment: If random samples of size 3 are drawn without replacement from the population consisting of four numbers 4, 5, 5, 7. Find the sample mean $\bar{X}$ for each sample and make a sampling distribution of $\bar{X}$. Calculate the mean and standard deviation of this sampling distribution. Compare your calculations with the population parameters.

Hint's: $\mu = \mu_{\bar{x}} = 5.25$ and $\sigma = \sigma_{\bar{x}} = 0.3632$.

Assignment: A population consists of the five numbers 2, 3, 6, 8, 11. Consider all possible samples of size two which can be drawn with replacement from this population. Find the following:

- (i) the mean of the population,
- (ii) the standard deviation of the population,
- (iii) the mean of the sampling distribution, and
- (iv) the standard deviation of the sampling distribution.

Hint's: $\mu = \mu_{\bar{x}} = 6$ and $\sigma = 3.29$ and $\sigma_{\bar{x}} = 2.32$.

1.5.2 Parent Distribution

Let we make a measurement x_i of a quantity X , we expect our observation to approximate the quantity, but we do not expect the experiment data point to be exactly equal to the quantity. In order to make more and more measurements, a pattern will emerge from the data. Some of the measurements will be too large, some will be too small. On the average, however, we expect them to be distributed around the correct value. If we could make an infinite number of measurements, then we could describe exactly the distribution of the data points. This is not possible in practice, but we can hypothesize the existence of such a distribution that determines the probability of obtaining a particular result in a single measurement. This distribution is called the *parent distribution*.